Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

University of Pisa, a.y. 2025-2026

Alessandro Gori*

Thesis for the Master's degree in Physics

Abstract

[To be continued. . .]

- Appendix on s-wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme
- Set ComputerModern font in all CairoMakie figures

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*a.gori23@studenti.unipi.it / nepero27178@github.com

List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ-wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued...]

Chapter 1

Theoretical introduction

[To be continued…]

1.1 Antiferromagnetic ordering in the Hubbard model

Consider the ordinary Hubbard model:

$$\hat{H} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \qquad t, U > 0 \tag{1.1}$$

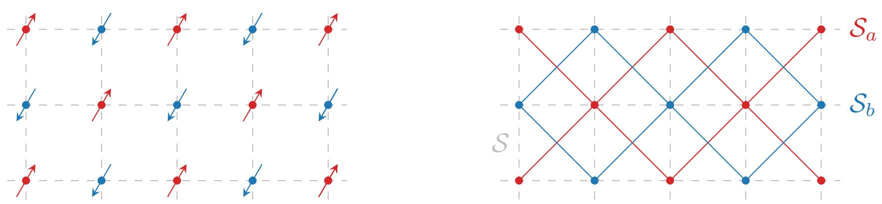
The two competing mechanisms are site-hopping of amplitude t and local repulsion of amplitude U. For this model defined on a bipartite lattice at half filling and fixed electron number, it is well known [**hirsch1985hubbard**] that, below a certain critical temperature T_c and above some (small) critical repulsion U_c/t, the ground-state acquires antiferromagnetic (AF) long-range ordering. schematically depicted in Fig. 1.1a. The mechanism for the formation of the AF phase takes advantage of virtual hopping, as described in App. **??**; the Mean-Field Theory (MFT) treatment of ferromagnetic-antiferromagnetic orderings in 2D Hubbard lattices is discussed in App. **??**.

In this chapter the discussion is limited to the two-dimensional square lattice Hubbard model. The lattice considered has L_ℓ sites on side $\ell = x, y$, thus a total of $L_x L_y$ sites. The total number of single electron states is given by $D = 2L_x L_y$. All theoretical discussion neglects border effects, thus considering $D \to +\infty$.

1.2 The Extended Fermi-Hubbard model

The Extended Fermi-Hubbard model is defined by:

$$\hat{H} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} \tag{1.2}$$



a | Antiferromagnetic phase.

b | Polarized sublattices.

Figure 1.1 | Schematic representation of the AF phase. Fig. 1.1a shows a portion of the square lattice with explicit representation of the spin for each site. Fig. 1.1b divides the square lattice $\mathcal{S}$ in two polarized sublattices $\mathcal{S}_a$, $\mathcal{S}_b$. The AF phase results from the interaction of two inversely polarized "ferromagnetic" square lattices.

Note that, on a square lattice, we can perform the summation over NN just as

$$\sum_{\langle ij\rangle} \equiv \sum_{i\in\mathcal{S}_a} \left[\delta_{j=i+\delta_x} + \delta_{j=i-\delta_x} + \delta_{j=i+\delta_y} + \delta_{j=i-\delta_y}\right]$$

where notation of Fig. 1.1b has been used. The last term represents an effective attraction between neighboring electrons, of amplitude V. Such an interaction is believed [**cao2025p-wave**] necessary to describe the insurgence of high-T_c superconductivity in cuprate SCs. [To be continued...]

1.2.1 Experimental insight on NN attraction

[Add:

- High T_c SC in cuprates;
- Experimental evidence of topological SC;
- Insertion of the non-local attraction;

]

Chapter 2

Introduction to Hartree-Fock methods

The Hartree-Fock (HF) method is a variational approach to determine the ground-state wavefunction of a many-body hamiltonian. The wavefunction is searched in a variational scheme as the one minimising energy, limiting search to a specific subspace of states. The general idea is to approximate the real hamiltonian with another non-interacting fermionic hamiltonian. The new operator should be quadratic in terms of fermionic operators.

First, we need to define what Hilbert subspace we work with. Then we need to identify a way of parametrising the possible non-interacting hamiltonians and their respective ground-states, and finally a method to choose the best approximation to our original model. This is done in next sections. The present discussion is largely based on [6, 8].

2.1 Prerequisites and generalities

Let us start by discussing the properties of the many-body states we wish to use to approximate the real ground-state of the system. We introduce Slater determinants to grasp the general idea behind HF variational search, and then extend to a larger set of states.

2.1.1 Slater determinants

A many-body wavefunction describing a collective electronic state needs to be antisymmetric under interchange of coordinates and spins $(\mathbf{r}_i, \sigma_i)$, $(\mathbf{r}_j, \sigma_j)$ of any couple of electrons. The simplest way to build a many-body wavefunction satisfying this rule is "to antisymmetrize" a set of orthonormal N one-body spin-orbitals, being N the total number of electrons and D the total number of spin-orbitals. Let $\{\psi_i\}$ be said set,

$$\frac{1}{2\mathcal{V}} \sum_{\sigma} \int \mathrm{d}\mathbf{r}\ \psi_i(\mathbf{r}, \sigma)\psi_j(\mathbf{r}, \sigma) = \delta_{ij}$$

being $\mathcal{V}$ the total volume. Let then the antisymmetrization operator

$$\mathcal{A} \equiv \frac{1}{\sqrt{N!}} \sum_{\mathrm{P} \in \mathrm{S}_N} (-1)^{\sigma(\mathrm{P})} \mathrm{P}$$

being S_N the $N!$ elements symmetric group of N objects permutations, and $\sigma(\mathrm{P})$ the "parity factor" of any permutation $\mathrm{P} \in \mathcal{S}_N$. $\sigma(\mathrm{P})$ counts the number of exchanges needed to achieve a given configuration. This way, an even number of exchanges gives $+1$, an odd number gives -1. The permutation operator exchanges spin-coordinates couples,

$$\mathrm{P} \quad : \quad (\mathbf{r}, \sigma) \leftrightarrow (\mathbf{r}', \sigma')$$

thus any given pair is exchanged with another.

Equipped with $\mathcal{A}$, we define the many-body wavefunction

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2; \cdots ; \mathbf{r}_N, \sigma_N) = \mathcal{A} \left\{ \prod_{i=1}^{N} \psi_i(\mathbf{r}_i, \sigma_i) \right\}$$

which by construction is antisymmetric under exchange of coordinates and spins pairs. This class of wavefunctions, obtained as antisymmetric combinations of product states, are representable as so-called Slater determinants,

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2; \cdots ; \mathbf{r}_N, \sigma_N) = \frac{1}{\sqrt{N!}} \det \begin{bmatrix} \psi_1(\mathbf{r}_1, \sigma_1) & \psi_1(\mathbf{r}_2, \sigma_2) & \cdots & \psi_1(\mathbf{r}_N, \sigma_N) \\ \psi_2(\mathbf{r}_1, \sigma_1) & \psi_2(\mathbf{r}_2, \sigma_2) & \cdots & \psi_2(\mathbf{r}_N, \sigma_N) \\ \vdots & \vdots & & \vdots \\ \psi_N(\mathbf{r}_1, \sigma_1) & \psi_N(\mathbf{r}_2, \sigma_2) & \cdots & \psi_N(\mathbf{r}_N, \sigma_N) \end{bmatrix}$$

Determinants of matrices with two equal rows or two equal columns are null. Thus, all spin-orbitals must differ, as well as no single spin-orbital ψ_i can be doubly occupied (which is, all spin-coordinate couples must be different). Pauli exclusion principle is included in this framework.

Any normalised combinations of Slater determinants gives an antisymmetric wavefunction, although not necessarily representable as a single Slater determinant. In HF theory, we search for the best possible choice of orthonormal one-body spin orbitals $\{\psi_1, \psi_2, \cdots, \psi_N\}$ to approximate the true ground-state of the model. The space of possible "Slater-like" many-body wavefunctions is the parametric space where we perform the variational constrained search.

The EHM is formulated in second-quantisation. It is then useful to represent Slater-like states in this language. Let us map all the spin-orbitals from the discussion above to a couple of annihilation-creation fermionic operators,

$$\psi_i \quad \rightarrow \quad \hat{c}_i, \hat{c}_i^\dagger$$

Any Slater state can be expressed in second-quantisation as:

$$|\Psi\rangle = \prod_{i=1}^{N} \hat{c}_i^\dagger \, |\Omega\rangle \tag{2.1}$$

being $|\Omega\rangle$ the vacuum.

2.1.2 Wick's theorem applied to Slater determinants

The concept of vacuum state needs to be interpreted in relation to an appropriate choice of second-quantisation annihilation and creation operators. For a given set of fermion operators, the vacuum is that state mapped to zero by any annihilation operator. As we will see, to correctly define the vacuum state is crucial in order to apply Wick's theorem and develop an HF approximation of EHM. Before moving to generalisation, let us discuss a playground setup.

Consider some unspecified model whose ground state at null temperature is given in Eq. (2.1) for a given set of electronic operators $\hat{c}_i$, $\hat{c}_i^\dagger$. Let $\hat{c}_i$ with $N < i < H$ identify all spin-orbitals not already included in the definition of $|\Psi\rangle$ of Eq. (2.1) (thus, non populated). Then it holds:

$$\begin{aligned} 0 \leq i \leq N \quad &: \quad \hat{c}_i^\dagger \, |\Psi\rangle = 0 \\ i > N \quad &: \quad \hat{c}_i \, |\Psi\rangle = 0 \end{aligned}$$

Let:

$$\hat{a}_i \equiv \begin{cases} \hat{c}_i^\dagger & \text{for } 0 \leq i \leq N \\ \hat{c}_i & \text{for } i > N \end{cases}$$

These new operators satisfy a Fermi algebra $\{\hat{a}_i, \hat{a}_j^\dagger\} = \delta_{ij}$. They are annihilation and creation operators above the "vacuum" state $|\Psi\rangle$. In other words, any Slater state like Eq. (2.1) can be seen as a vacuum state for a suitably chosen set of fermionic operators.

This property is handy for evaluation of expectation values (EV). Let $\hat{O}^{(\ell)}$ be a linear combination of $\hat{c}_i$ or $\hat{c}_i^\dagger$,

$$\hat{O}^{(\ell)} \equiv \sum_{i=1}^{D} \left(u_i^{(\ell)} \hat{c}_i + v_i^{(\ell)} \hat{c}_i^\dagger \right) \tag{2.2}$$

for some coefficients $\left\{ u_i^{(\ell)}, v_i^{(\ell)} \right\}$. Consider the product operator:

$$\hat{O} \equiv \hat{O}^{(1)} \hat{O}^{(2)} \cdots \hat{O}^{(2k)}$$

for $k \in \mathbb{N}$. Inserting the above decomposition, we get a large sum of many terms, each containing some sequence of annihilations and creations. Using linearity we can concentrate on one specific term of these, derive results and then recompose them for the original operator. Consider one of these terms,

$$\hat{\mathcal{O}} = \cdots + (\text{coefficient}) \times \hat{C}_1\hat{C}_2 \cdots \hat{C}_{2k} + \cdots$$

with $\hat{C}_i$ some fermionic operator, either $\hat{c}_j$ or $\hat{c}_j^\dagger$ for some j. Now, Wick's theorem states that

$$\begin{aligned}
\hat{C}_1\hat{C}_2 \cdots \hat{C}_{2k} = \mathrm{N}\left\{\hat{C}_1\hat{C}_2 \cdots \hat{C}_{2k}\right\} \\
&+ \sum_{(\alpha\beta)} (-1)^{\zeta_{\alpha\beta}} D_{\alpha\beta} \mathrm{N}\left\{\prod_{i\neq\alpha,\beta} \hat{C}_i\right\} \\
&+ \sum_{(\alpha\beta)(\gamma\delta)} (-1)^{\zeta_{\alpha\beta}} D_{\alpha\beta} (-1)^{\zeta_{\gamma\delta}} D_{\gamma\delta} \mathrm{N}\left\{\prod_{i\neq\alpha,\beta,\gamma,\delta} \hat{C}_i\right\} \\
&+ \cdots
\end{aligned}$$

where $\mathrm{N}\{\cdots\}$ indicates normal ordering, $D_{\alpha\beta}$ is the contraction of the pair $\hat{C}_\alpha\hat{C}_\beta$ and $\zeta_{\alpha\beta}$ is a symmetry factor included to take care of the number of fermionic algebra. In this context contractions are defined as expectation values over $|\Psi\rangle$. Sums are performed over pairs of indices, implicitly taken to be different from each other. To take averages of this last operator over $|\Psi\rangle$ is a not-so-hard task, given the simple expression of the state in terms of electronic annihilations and creations. However it can be made simpler: if we change basis and express all operators in terms of $\hat{a}$, $\hat{a}^\dagger$,

$$\hat{C}_i \quad \rightarrow \quad \hat{A}_i$$

the above decomposition holds identically, with slight change of notation. Now, taking a zero temperature average on $|\Psi\rangle$ (the vacuum with respect to the new operators) all normal-ordered terms disappear,

$$\langle\Psi|\mathrm{N}\{\hat{A}_i\hat{A}_j\cdots\}|\Psi\rangle = 0$$

thus in the decomposition above only fully contracted terms survive. Going back to $\hat{c}$ operators, we can re-express fully contracted terms in the original basis. Thus:

$$\left\langle \hat{C}_1\hat{C}_2 \cdots \hat{C}_{2k} \right\rangle = \sum (\text{total sign factor}) \times (\text{product of complete contractions})$$

where sum is intended, implicitly, over all possible ways of choosing k couples of indices from a starting set of $2k$ elements.

Taking another step back, due to linearity we can apply this result to all terms of $\mathcal{O}$ and conclude that the EV of $\mathcal{O}$ decomposes as a sum of 2-points EVs. **This is the defining property of gaussian states**. In this procedure, the only key property of Slater states we used was the fact that they are vacuum to some fermionic basis. Finite-temperature extension of Wick's theorem as well as more complicated change of basis are possible, but the concept remains the same: on a Slater state, which is essentially a non-interacting state for some set of fermions, Wick's theorem is particularly well-behaved.

From the point above we can conclude that:

$$\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\gamma \hat{c}_\delta \rangle = \langle \hat{c}_\alpha^\dagger \hat{c}_\delta \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\gamma \rangle - \langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \langle \hat{c}_\gamma^\dagger \hat{c}_\delta \rangle$$

being the average intended on a Slater state. In this context, only contractions with an equal number of creation and annihilation operators contribute.

2.1.3 Generalisation to Bogoliubov-Valatin states

In HF theory we look for a good approximation of the true many-body wavefunction constraining search to a class of states. Introducing Slater determinants was useful to grasp the concrete idea behind the method. **However, in order to treat some physical phenomena we need to**

extend the variational procedure to a other sets of states. For example, Bardeen-Cooper-Shrieffer (BCS) theory of superconductivity gives a many-body ground-state which is not representable as a Slater determinant of "standard" electronic operators. In this section we extend the notions introduced before and set grounds for the generalised Hartree-Fock-Bogoliubov (HFB) method we use in this thesis.

Consider the Nambu spinor

$$\hat{\Phi} \equiv \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix}$$

being $\hat{\mathbf{c}}$ the vector of all annihilation operators, ordered, and $\hat{\mathbf{c}}^\dagger$ of all creation operators. Let us assume the vectors have D entries. Consider an unitary map M, such that

$$\hat{\Gamma} \equiv M\hat{\Phi} \quad \text{being} \quad \hat{\Gamma} \equiv \begin{bmatrix} \hat{\gamma} \\ \hat{\gamma}^\dagger \end{bmatrix}$$

The new Nambu vectors $\hat{\gamma}$ and $\hat{\gamma}^\dagger$ contain our new, generalised fermionic operators in second quantisation. Let:

$$M \equiv \begin{bmatrix} U & V \\ U^\dagger & V^\dagger \end{bmatrix}$$

If M is unitary, it follows:

$$\begin{aligned}
\mathbb{I}_{2D\times 2D} &= MM^\dagger \\
&= \begin{bmatrix} U & V \\ U^\dagger & V^\dagger \end{bmatrix} \begin{bmatrix} U & U^\dagger \\ V^\dagger & V \end{bmatrix} \\
&= \begin{bmatrix} UU^\dagger + VV^\dagger & U^2 + V^2 \\ \left(U^\dagger\right)^2 + \left(V^\dagger\right)^2 & U^\dagger U + V^\dagger V \end{bmatrix}
\end{aligned}$$

thus, among other conditions, $UU^\dagger + VV^\dagger = \mathbb{I}_{D\times D}$.

The new operators in $\hat{\Gamma}$ satisfy Fermi anticommutation rules. In fact, writing the mapping explicitly for any of the new operators and some index $i \in [1, D]$,

$$\hat{\gamma}_i \equiv \sum_{j=1}^{D} U_{ij}\hat{c}_j + \sum_{j=1}^{D} V_{ij}\hat{c}_j^\dagger \qquad \hat{\gamma}_i^\dagger \equiv \sum_{j=1}^{D} U_{ji}^*\hat{c}_j^\dagger + \sum_{j=1}^{D} V_{ji}^*\hat{c}_j$$

and substituting in the anti-commutator

$$\begin{aligned}
\{\hat{\gamma}_i, \hat{\gamma}_j^\dagger\} &= \sum_{i'=1}^{D} U_{ii'} \sum_{j'=1}^{D} U_{j'j}^* \{\hat{c}_{i'}, \hat{c}_{j'}^\dagger\} + \sum_{i'=1}^{D} V_{ii'} \sum_{j'=1}^{D} V_{j'j}^* \{\hat{c}_{i'}^\dagger, \hat{c}_{j'}\} \\
&= \sum_{i'=1}^{D} U_{ii'} \sum_{j'=1}^{D} U_{j'j}^* \delta_{i'j'} + \sum_{i'=1}^{D} V_{ii'} \sum_{j'=1}^{D} V_{j'j}^* \delta_{i'j'} \\
&= \left[UU^\dagger + VV^\dagger\right]_{ij} \\
&= \delta_{ij}
\end{aligned}$$

being M unitary.

Assume now $|\Psi\rangle$ to be the vacuum state for the new operators,

$$\forall i \quad : \quad \hat{\gamma}_i |\Psi\rangle = 0$$

Suppose we want to evaluate, with identical meaning of the notation as in previous section, something like

$$\left\langle \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} \right\rangle$$

We can apply the result obtained above. Indeed, inverse maps linking old and new operators read:

$$\hat{c}_i \equiv \sum_{j=1}^{D} U_{ij}\hat{\gamma}_j + \sum_{j=1}^{D} U_{ji}^*\hat{\gamma}_j^\dagger \qquad \hat{c}_i^\dagger \equiv \sum_{j=1}^{D} V_{ji}^*\hat{\gamma}_j^\dagger + \sum_{j=1}^{D} V_{ij}\hat{\gamma}_j$$

In Sec. 2.1.2 we argued that, when averaging a product of operators like the one of Eq. (2.2), only 2-points EVs contribute. This time the idea is the same, substituting $\mathcal{O}^{(\ell)} \to \hat{C}_\ell$ and $\hat{c}_i, \hat{c}_i^\dagger \to \hat{\gamma}_i, \hat{\gamma}_i^\dagger$. Thus, also here we can decompose the product operator, apply Wick's theorem, discard all non-fully-contracted terms and recompose the results. All we used was just the unitary mapping between $\hat{\Phi}$ and $\hat{\Gamma}$, the fact that $\gamma_i, \hat{\gamma}_i^\dagger$ obey Fermi anticommutation rules and "see" $|Psi\rangle$ as a vacuum state.

As in Sec. 2.1.2, we conclude that:

$$\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\gamma \hat{c}_\delta \rangle = \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\delta \rangle \langle \hat{c}_\beta^\dagger \hat{c}_\gamma \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \langle \hat{c}_\gamma^\dagger \hat{c}_\delta \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \rangle \langle \hat{c}_\gamma \hat{c}_\delta \rangle}_{\text{Cooper}}$$

The subscripts indicate the name we will give to these terms throughout the text. Note that, differently from "pure Slater states" EVs with two annihilations or two creations do not vanish, in general.

All of this holds when averaging is performed on a state, $|\Psi\rangle$, which is vacuum for a set of operators linked to fermionic Nambu spinor $\hat{\Phi}$ by some unitary operation. This set of states is the domain of our variational constrained search. This kind of unitary mapping is often referred to as Bogoliubov-Valatin transformation, and the method hereby presented as Hartree-Fock-Bogoliubov (HFB). For simplicity we will stick, from now on, to the name "HF method" referring to this discussion. **This argument has been carried out at zero temperature, but can be extended to finite temperature using appropriate path-integral techniques**.

2.2 Sketch of HF variational method

Consider the generic quartic hamiltonian:

$$\hat{H} = \sum_{\alpha,\beta} A_{\alpha\beta} \hat{c}_\alpha^\dagger \hat{c}_\beta + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \hat{c}_\alpha^\dagger \hat{c}_\beta^\dagger \hat{c}_\gamma \hat{c}_\delta$$

where

$$A_{\alpha\beta} = A_{\beta\alpha}^* \qquad B_{\alpha\beta\gamma\delta} = B_{\delta\gamma\beta\alpha}^* \tag{2.3}$$

to ensure hermiticity. Moreover, the model must show the following symmetry:

$$B_{\alpha\beta\gamma\delta} = B_{\beta\alpha\delta\gamma} \tag{2.4}$$

because exchanging the position of the two annihilations and the two creations, the operator must remain the same.

Our aim is to approximate the above hamiltonian with something like:

$$\hat{H} \simeq \hat{H}_{\text{HF}} \equiv \sum_{\alpha,\beta} k_{\alpha\beta} \hat{\gamma}_\alpha^\dagger \hat{\gamma}_\beta$$

Old and new operators must be linked by an unitary transform: with identical notation as in previous sections, $\hat{\Gamma} = M\hat{\Phi}$. To find the best approximation to the original hamiltonian means to identify both the γ operators and the k coefficients.

2.2.1 Properties of the target quadratic hamiltonian

Let us enumerate briefly some properties of the target hamiltonian. We will use also the following notation:

$$\hat{a}_\alpha^- \equiv \hat{c}_\alpha \qquad \hat{a}_\alpha^+ \equiv \hat{c}_\alpha^\dagger$$

Due to linearity of the unitary mapping, $\hat{H}_{\text{HF}}$ can be expressed also as the most general quadratic hamiltonians for electronic operators:

$$\hat{H}_{\text{HF}} = \sum_{\alpha,\beta} \sum_{p,q} h_{\alpha\beta}^{pq} \hat{a}_\alpha^p \hat{a}_\beta^q$$

We moved all our ignorance into the new $h^{pq}_{\alpha\beta}$ fields.

We can infer some mathematical properties of the fields $h^{pq}_{\alpha\beta}$. First, note that the hamiltonian can be represented through Nambu spinors as

$$\hat{H}_{\text{HF}} = \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix}^\dagger \begin{bmatrix} h^{+-} & h^{++} \\ h^{--} & h^{-+} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix}$$

From hamiltonian hermiticity we conclude that:

$$\hat{H}^\dagger_{\text{HF}} = \hat{H}_{\text{HF}} \qquad \Longrightarrow \qquad h^{++} = \left[h^{--}\right]^\dagger$$

which reads, element-wise:

$$h^{++}_{\alpha\beta} = \left(h^{--}_{\beta\alpha}\right)^* \tag{2.5}$$

Same-sign off-diagonal fields are linked.

Moreover, any hamiltonian can be arbitrarily chosen to be traceless: Physics is not affected by constant energy shifts. Thus we can set

$$\operatorname{Tr} h^{+-} + \operatorname{Tr} h^{-+} = 0$$

Our variational argument does not depend neither on the trace of the fields nor the trace of the original hamiltonian. From this we conclude that also on-diagonal terms are linked: indeed, considering the explicit form of the hamiltonian:

$$\begin{aligned}
\hat{H}_{\text{HF}} &= \sum_{\alpha,\beta} \left[h^{+-}_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}_\beta + h^{-+}_{\alpha\beta} \hat{c}_\alpha \hat{c}^\dagger_\beta\right] + (\text{same sign terms}) \\
&= \sum_{\alpha,\beta} \left[h^{+-}_{\alpha\beta} \left(\delta_{\alpha\beta} - \hat{c}_\beta \hat{c}^\dagger_\alpha\right) + h^{-+}_{\alpha\beta} \left(\delta_{\alpha\beta} - \hat{c}^\dagger_\beta \hat{c}_\alpha\right)\right] + (\text{same sign terms}) \\
&= -\sum_{\alpha,\beta} \left[h^{+-}_{\beta\alpha} \hat{c}_\alpha \hat{c}^\dagger_\beta + h^{-+}_{\beta\alpha} \hat{c}^\dagger_\alpha \hat{c}_\beta\right] + (\text{same sign terms})
\end{aligned}$$

In second row we got rid of the traces sum and exchanged indices $\alpha \leftrightarrow \beta$. Confronting last and first row we conclude:

$$h^{+-}_{\beta\alpha} = -h^{-+}_{\alpha\beta} \qquad h^{-+}_{\beta\alpha} = -h^{+-}_{\alpha\beta} \tag{2.6}$$

thus, also on-diagonal fields are linked. We are allowed to determine, say, just h^{+-}: its counterpart h^{-+} is completely determined.

2.2.2 Self-consistent variational solution

We follow and extend the variational scheme proposed by Fabrizio [6]. Let:

$$\Delta^{pq}_{\alpha\beta} \equiv \left\langle \hat{a}^p_\alpha \hat{a}^q_\beta \right\rangle$$

thus, explicitly

$$\Delta^{+-}_{\alpha\beta} = \left\langle \hat{c}^\dagger_\alpha \hat{c}_\beta \right\rangle \qquad \Delta^{-+}_{\alpha\beta} = \left\langle \hat{c}_\alpha \hat{c}^\dagger_\beta \right\rangle \qquad \Delta^{++}_{\alpha\beta} = \left\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \right\rangle \qquad \Delta^{--}_{\alpha\beta} = \left\langle \hat{c}_\alpha \hat{c}_\beta \right\rangle \tag{2.7}$$

The first two are "normal densities"; the second two are "anomalous densities". Conjugation of the last two gives a symmetry relation similar to Eq. (2.5):

$$\left(\Delta^{++}_{\alpha\beta}\right)^\dagger = \Delta^{--}_{\alpha\beta} \tag{2.8}$$

The HF ground-state energy is given by:

$$\begin{aligned}
E_{\text{HF}} &= \left\langle \hat{H}_{\text{HF}} \right\rangle \\
&= \sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \Delta^{pq}_{\alpha\beta}
\end{aligned}$$

Here $\Delta^{pq}_{\alpha\beta}$ are functions of the fields $h^{pq}_{\alpha\beta}$. Derive energy:

$$\begin{aligned}\frac{\partial E_{\mathrm{HF}}}{\partial h^{rs}_{\mu\nu}} &= \frac{\partial}{\partial h^{rs}_{\mu\nu}} \sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \Delta^{pq}_{\alpha\beta} \\ &= \Delta^{rs}_{\mu\nu} + \sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \frac{\partial \Delta^{pq}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\end{aligned}$$

From Hellman-Feynman theorem [8], we know that for a parametric hamiltonian it holds:

$$\begin{aligned}\frac{\partial E_{\mathrm{HF}}}{\partial h^{rs}_{\mu\nu}} &= \langle\Psi| \frac{\partial \hat{H}_{\mathrm{HF}}}{\partial h^{rs}_{\mu\nu}} |\Psi\rangle \\ &= \langle\Psi| \frac{\partial}{\partial h^{rs}_{\mu\nu}} \sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \hat{a}^p_\alpha \hat{a}^q_\beta |\Psi\rangle \\ &= \langle \hat{a}^r_\mu \hat{a}^s_\nu \rangle \\ &= \Delta^{rs}_{\mu\nu}\end{aligned}$$

The last two equations imply:

$$\sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \frac{\partial \Delta^{pq}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} = 0 \tag{2.9}$$

This must hold for any possible choice of fields $h^{pq}_{\alpha\beta}$. Note that we did not impose the energy derivative to be zero: it is not the HF energy we are minimising.

Let us get back to the actual hamiltonian $\hat{H}$. To reduce it to a quadratic form is equivalent to assume that its ground-state has the properties described in previous sections. Then the following decomposition must hold:

$$\begin{aligned}E &= \langle \hat{H} \rangle \\ &= \sum_{\alpha,\beta} A_{\alpha\beta} \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta \rangle \\ &= \sum_{\alpha,\beta} A_{\alpha\beta} \langle \hat{a}^+_\alpha \hat{a}^-_\beta \rangle + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \left(\langle \hat{a}^+_\alpha \hat{a}^+_\beta \rangle \langle \hat{a}^-_\gamma \hat{a}^-_\delta \rangle - \langle \hat{a}^+_\alpha \hat{a}^-_\gamma \rangle \langle \hat{a}^+_\beta \hat{a}^-_\delta \rangle + \langle \hat{a}^+_\alpha \hat{a}^-_\delta \rangle \langle \hat{a}^+_\beta \hat{a}^-_\gamma \rangle \right) \\ &= \sum_{\alpha,\beta} A_{\alpha\beta} \Delta^{+-}_{\alpha\beta} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \left(\Delta^{++}_{\alpha\beta} \Delta^{--}_{\gamma\delta} - \Delta^{+-}_{\alpha\gamma} \Delta^{+-}_{\beta\delta} + \Delta^{+-}_{\alpha\delta} \Delta^{+-}_{\beta\gamma} \right) \\ &= \sum_{\alpha,\beta} A_{\alpha\beta} \Delta^{+-}_{\alpha\beta} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} \left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta} \right) \Delta^{+-}_{\alpha\beta} \Delta^{+-}_{\gamma\delta} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \Delta^{++}_{\alpha\beta} \Delta^{--}_{\gamma\delta}\end{aligned}$$

In second passage we inserted the the relations (2.7). Take the derivative:

$$\begin{aligned}\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta} A_{\alpha\beta} \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} \left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta} \right) \left(\frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \Delta^{+-}_{\gamma\delta} + \Delta^{+-}_{\alpha\beta} \frac{\partial \Delta^{+-}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}} \right) \\ + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \left(\frac{\partial \Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \Delta^{--}_{\gamma\delta} + \Delta^{++}_{\alpha\beta} \frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}} \right)\end{aligned}$$

Using Eq. (2.4) we end up with:

$$\begin{aligned}\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta} \left[A_{\alpha\beta} + \sum_{\gamma,\delta} \left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta} \right) \Delta^{+-}_{\gamma\delta} \right] \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \\ + \sum_{\alpha,\beta,\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta} \Delta^{--}_{\gamma\delta}}{2} \frac{\partial \Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + \sum_{\alpha,\beta,\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta} \Delta^{++}_{\alpha\beta}}{2} \frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}\end{aligned}$$

Let now:

$$C_{\alpha\beta} \equiv \frac{1}{2}\left[A_{\alpha\beta} + \sum_{\gamma,\delta}\left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\Delta^{+-}_{\gamma\delta}\right] \qquad D_{\alpha\beta} \equiv \sum_{\gamma,\delta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{--}_{\gamma\delta}}{2} \tag{2.10}$$

From Fermi anticommutation rules, we have:

$$\Delta^{+-}_{\alpha\beta} = \delta_{\alpha\beta} - \Delta^{-+}_{\beta\alpha} \quad \Longrightarrow \quad \frac{\partial\Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} = -\frac{\partial\Delta^{-+}_{\beta\alpha}}{\partial h^{rs}_{\mu\nu}}$$

which implies:

$$\begin{aligned}\sum_{\alpha,\beta} 2C_{\alpha\beta}\frac{\partial\Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} &= \sum_{\alpha,\beta}\left(C_{\alpha\beta}\frac{\partial\Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + C_{\alpha\beta}\frac{\partial\Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\right)\\ &= \sum_{\alpha,\beta}\left(C_{\alpha\beta}\frac{\partial\Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} - C_{\beta\alpha}\frac{\partial\Delta^{-+}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\right)\end{aligned}$$

Moreover, using Eqns. (2.3), (2.8),

$$\sum_{\alpha,\beta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}}{2} = \sum_{\alpha,\beta}\left(\frac{B_{\delta\gamma\beta\alpha}\Delta^{--}_{\beta\alpha}}{2}\right)^*$$

Using these two relations, we conclude:

$$\sum_{\alpha,\beta,\gamma,\delta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{++}_{\alpha\beta}}{2}\frac{\partial\Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}} = \sum_{\gamma,\delta} D^*_{\delta\gamma}\frac{\partial\Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}$$

Recollect everything and impose the solution to be a stationary point of the energy,

$$\frac{\partial E}{\partial h^{rs}_{\mu\nu}} \stackrel{!}{=} 0$$

The notation "$\stackrel{!}{=}$" indicates, throughout the thesis, that we are imposing some condition. In the end we have:

$$\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta}\left(C_{\alpha\beta}\frac{\partial\Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} - C_{\beta\alpha}\frac{\partial\Delta^{-+}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + D_{\alpha\beta}\frac{\partial\Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + D^*_{\beta\alpha}\frac{\partial\Delta^{--}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}}\right) \stackrel{!}{=} 0 \tag{2.11}$$

Now compare Eqns. (2.9) and (2.11). We recognise $h^{+-}_{\alpha\beta} = C_{\alpha\beta}$ and $h^{++}_{\alpha\beta} = D_{\alpha\beta}$; the remaining fields are determined by the symmetry relations (2.6) and (2.5). Explicitly, the fields are self-consistently determined by the following two equations:

$$h^{+-}_{\alpha\beta} = \frac{1}{2}\left[A_{\alpha\beta} + \sum_{\gamma,\delta}\left(B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta}\right)\Delta^{+-}_{\gamma\delta}\right] \tag{2.12}$$

$$h^{++}_{\alpha\beta} = \sum_{\gamma,\delta}\frac{B_{\alpha\beta\gamma\delta}\Delta^{--}_{\gamma\delta}}{2} \tag{2.13}$$

The right-hand side of both equations depends on the fields themselves: $\Delta^{pq}_{\alpha\beta}$ are defined as expectation values of two-points fermionic operators over the ground-state $|\Psi\rangle$. For a given set of fields $h^{pq}_{\alpha\beta}$, the latter is determined (up to some unitary transformation) as the ground-state of a fermionic non-interacting model. In other words, being $\mathbf{h}$ the tensor containing all fields,

$$\Delta^{pq}_{\alpha\beta}[\mathbf{h}] = \langle\Psi[\mathbf{h}]|\hat{a}^p_\alpha\hat{a}^q_\beta|\Psi[\mathbf{h}]\rangle \tag{2.14}$$

This set of equations defines our computational problem. We are searching a set of parameters $\{\Delta^{pq}_{\alpha\beta}\}$ such that, determining the fields via Eqns. (2.12), (2.13), then Eq. (2.14) is respected. **The**

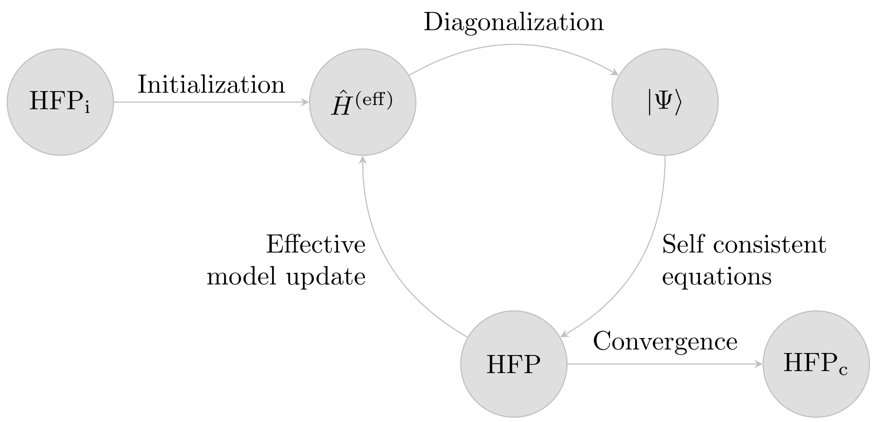


Figure 2.1 Schematization of the self-consistent Hartree Fock method described in Sec. ??. The effective model is solved, the solution is used to estimate self-consistently the Hartree Fock vector, which is then used to update iteratively the model itself. $\mathrm{HFP_i}$ indicates the initialzer HFPs vector, $\mathrm{HFP_c}$ the converged HFPs vector.

great advantage of this scheme is that, with an appropriate choice of the Hilbert space basis (for example, moving to reciprocal space for a translationally-invariant model), we can implement easily model symmetries.

The list of parameters $\{\Delta^{pq}_{\alpha\beta}\}$ to be determined self-consistently may be not-so-short: by looking at Eq. (2.14), the indices α, β, p, q are spanning a large Hilbert space. However, this equation can be manipulated and transformed, reducing the set of parameters to be determined self-consistently to a very manageable number. For each phase we define these last as "Hartree-Fock parameters" (HFPs): a set of objects self-consistently determined by a set equivalent, but transformed, version of Eq. (2.14).

2.3 Sketch of the iHF algorithm

In this section we give a practical sketch of the algorithm we use to compute Eq. (2.14) for all given phases. We refer to this technique as "iterative Hartree Fock" (iHF). Let $\mathbf{v}$ be the vector containing all HFPs. Suppose we have N HFPs v_i: then

$$\mathbf{v} \in \mathbb{C}^N \qquad \mathbf{v} \equiv \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix}$$

In order to get a physical solution, each HFP must satisfy a self-consistency equation similar to Eq. (2.14). Each equation has the structure:

$$v_i = (\text{Some non-linear function of } \mathbf{v})$$

Let A be the non-linear operator collecting all these functions. The problem of finding a self-consistent solution can be reformulated as follows: $\mathbf{v}$ represents a physical solution if it is a fixed point of the map A,

$$\mathrm{A}\colon \mathbb{C}^N \to \mathbb{C}^N \qquad \mathrm{A}(\mathbf{v}) = \mathbf{v} \quad \Longrightarrow \quad \mathbf{v} \text{ is a solution.}$$

We implement a rather easy scheme: first, we obtain an analytic expression for A. Then we choose an initializer $\mathbf{v}_0$ and apply A to it, obtaining a new HFPs vector. Iterating this procedure, we "follow" the non-linear map A until we claim to have converged. Let us explain in detail:

1. Start with a generic quartic hamiltonian:

$$\hat{H} = \sum_{\alpha\beta} A_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}_\beta + \sum_{\alpha\beta\gamma\delta} B_{\alpha\beta\gamma\delta} \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta$$

Choose the model parameters and set an electronic density;

2. Following the discussion of Sec. 2.2, approximate analytically $\hat{H}$ with a quadratic hamiltonian

$$\hat{H} \simeq \hat{H}_{\rm HF} \equiv \sum_{\alpha\beta} \left[h^{+-}_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}_\beta - h^{+-}_{\beta\alpha} \hat{c}_\alpha \hat{c}^\dagger_\beta + h^{++}_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta + \left(h^{++}_{\beta\alpha} \right)^* \hat{c}_\alpha \hat{c}_\beta \right]$$

with the fields given by Eqns. (2.12), (2.13);

3. Impose selection rules on the fields $h^{+-}_{\alpha\beta}$, $h^{++}_{\alpha\beta}$, eventually eliminating those not respecting the symmetry sector we are working in. As an example: if we choose not to break charge conservation, set $\forall \alpha, \beta$: $h^{++}_{\alpha\beta} = 0$;

4. Choose a starting value for the HFPs vector $\mathbf{v}_0$;

5. Set $\mathbf{v} = \mathbf{v}_0$ and enter the iterative scheme of Fig. 2.1:

   a) Get a numeric approximation of the hamiltonian for the current value of $\mathbf{v}$, $\hat{H}_{\rm HF}[\mathbf{v}]$;
   b) Determine the value $\mu[\mathbf{v}]$ giving the desired density;
   c) Diagonalise the effective hamiltonian $\hat{H}_{\rm HF}[\mathbf{v}] - \mu[\mathbf{v}]\hat{N}$, and obtain the approximated ground-state $|\Psi[\mathbf{v}]\rangle$;
   d) Compute the new HFPs vector $\mathbf{v}$ using self-consistency equations;
   e) If the HFPs vector converged to a fixed point, exit; otherwise, iterate.

We voluntarily left unspecified the criterion we use for claiming that $\mathbf{v}$ "has converged". It will be specified later.

2.3.1 Fixed density simulations and the role of μ

In this thesis all derivations and simulations are carried out at fixed density. This requires a way to fix μ during iterations of the algorithm. This must be done iteratively: changing at each step the HFPs vector, we are actively modifying the approximate hamiltonian. Chemical potential must be determined accordingly to maintain the required density.

For a given choice of HFPs vector and chemical potential, the ground state $|\Psi[\mathbf{v}, \mu]\rangle$ is uniquely determined; then we compute density as

$$n(\mathbf{v}, \mu) \equiv \frac{1}{\mathcal{V}} \sum_\alpha \langle \Psi[\mathbf{v}, \mu] | \hat{c}^\dagger_\alpha \hat{c}_\alpha | \Psi[\mathbf{v}, \mu] \rangle$$

being $\mathcal{V}$ the volume. Suppose we fix density to a value $n_0 \in [0, 1]$. At step i, naming the current HFPs vector $\mathbf{v}_i$, μ_i (the chemical potential at current step) is determined as

$$\mu_i \equiv \min_\mu |n(\mathbf{v}_i, \mu) - n_0| \tag{2.15}$$

The density n must be a monotonically increasing function of μ: then we can translate the above equation, which would inspire a minimisation procedure, to a much simpler node-search rule:

$$\mu_i = \mu \mid_{\delta n(\mathbf{v}_i, \mu) = 0} \quad \text{being} \quad \delta n(\mathbf{v}_i, \mu) \equiv n(\mathbf{v}_i, \mu) - n_0$$

In order to determine μ_i we use a basic bisection algorithm:

1. Set an arbitrary tolerance $\Delta n > 0$ and a shifting parameter $\Delta\mu > 0$;

2. Choose starting lower and upper bounds $\mu_{\rm L} < \mu_{\rm U}$;

3. Compute $\delta n(\mathbf{v}_i, \mu_{\rm L})$. Evaluate:

   - if $|\delta n(\mathbf{v}_i, \mu_{\rm L})| \le \Delta n$, set $\mu_i = \mu_{\rm L}$ and halt;
   - if $\delta n(\mathbf{v}_i, \mu_{\rm L}) < -\Delta n$ do nothing;

- If instead $\delta n(\mathbf{v}_i, \mu_\mathrm{L}) > \Delta n$, shift the lower bound down, $\mu_\mathrm{L} \to \mu_\mathrm{L} - \Delta\mu$ and repeat from step 3;

4. Compute $\delta n(\mathbf{v}_i, \mu_\mathrm{U})$. Evaluate:

   - if $|\delta n(\mathbf{v}_i, \mu_\mathrm{U})| \leq \Delta n$, set $\mu_i = \mu_\mathrm{U}$ and halt;
   - if $\delta n(\mathbf{v}_i, \mu_\mathrm{U}) > \Delta n$ do nothing;
   - If instead $\delta n(\mathbf{v}_i, \mu_\mathrm{U}) < -\Delta n$, shift the upper bound up, $\mu_\mathrm{U} \to \mu_\mathrm{U} + \Delta\mu$ and repeat from step 4;

5. At this point, the the lower bound underestimates density while the upper bound overestimates. Start iterations:

   a) Compute the midpoint,
   $$\mu_\mathrm{M} = \frac{\mu_\mathrm{L} + \mu_\mathrm{U}}{2}$$

   b) Evaluate:
      - If $\delta n(\mathbf{v}_i, \mu_\mathrm{M}) \leq \Delta n$, set $\mu_i = \mu_\mathrm{M}$ and halt;
      - If $\delta n(\mathbf{v}_i, \mu_\mathrm{M}) > \Delta n$, set $\mu_\mathrm{U} = \mu_\mathrm{M}$ and repeat from point a);
      - If $\delta n(\mathbf{v}_i, \mu_\mathrm{M}) < -\Delta n$, set $\mu_\mathrm{L} = \mu_\mathrm{M}$ and repeat from point a);

[To be continued...]

Hence, if we are able to perform averages and know both $\mathbf{v}_i$ and n, at step i we infer the correct chemical potential. Now, as we will discuss $\tilde{\mu}$ will turn out to be an RMP itself, meaning that theoretically
$$\mu \to \mu + F[\mathbf{v}] \equiv \tilde{\mu}[\mathbf{v}]$$
being F some functional of the HFPs. In other words, a common effect of interactions under MFT will be precisely to shift the chemical potential. At step i, keeping the density fixed, the chemical potential we are determining numerically and using in our equations is then $\tilde{\mu}_i$. In simpler words, the designed algorithm "does not know" about chemical potential renormalization and thus just determines the number that solves Eq. (2.15). But we know that this number is made of two contributions: the "physical" μ_i and the effective shift produced by the model itself, $F[\mathbf{v}_i]$. So, when it comes to μ_i, this consideration becomes important.

It is the case of the ground state energy estimation. Let us abstract from the EHM and MFT, and consider a generic hamiltonian $\hat{H}$. To find the ground state $|\Psi\rangle$ means precisely to solve
$$\frac{\delta}{\delta \langle\Psi|} \langle\Psi|\hat{H}|\Psi\rangle = 0$$
Now, working with a fixed particles number N means we are performing a constrained minimization (which is, taking the functional derivative) on the composite operator
$$\hat{H} - \mu(\hat{N} - N) \equiv \hat{K} + \mu N$$
being μ the Lagrange multiplier, $\hat{N}$ the number operator and $\hat{K} \equiv \hat{H} - \mu\hat{N}$. Finding the constrained ground state parametrized by a vector $\mathbf{v}$ now means to solve
$$\mathbf{v} \implies |\Psi[\mathbf{v}]\rangle \quad : \quad \begin{cases} \dfrac{\delta}{\delta\langle\Psi|} \langle\Psi|\hat{K} + \mu N|\Psi\rangle \Big|_{|\Psi\rangle = |\Psi[\mathbf{v}]\rangle} = 0 \\ \langle\Psi[\mathbf{v}]|\hat{N}|\Psi[\mathbf{v}]\rangle = N \end{cases}$$
In the first line μN is derived to zero. The constraint is imposed in the second line. This means that, when it comes to determining the parametric ground state, $\hat{K}$ is functionally minimized. In this context μ (the Lagrange multiplier) sums with other effective terms arisen from algebraic manipulations of $\hat{H}$ giving a grand total of $\tilde{\mu}$. The number operator $\hat{N}$ average is performed over

the many-body ground state $|\Psi[\mathbf{v}]\rangle$ with the total effective chemical potential being $\tilde{\mu}$, not μ. However, when it comes to computing energy,

$$\begin{aligned}\langle \hat{H} \rangle &= \langle \Psi[\mathbf{v}]|\hat{K}|\Psi[\mathbf{v}]\rangle + \mu N \\ &= \Omega[\mathbf{v}] + (\tilde{\mu}[\mathbf{v}] - F[\mathbf{v}])N\end{aligned}$$

being Ω the expectation value (EV) of $\hat{K}$. In the last passage we wrote everything in terms of functionals of the HFPs vector $\mathbf{v}$. Using this relation, when the algorithm has converged, we can estimate the physical energy.

2.3.2 The reject-mix-iterate convergence scheme

Up to now, we did not specify when we consider the algorithm to "have converged". From a purely theoretical point of view, considering the set of self-consistency equations provided by MFT as a nonlinear map in HFPs space, we consider the run finished when our algorithm, following a non-trivial path in HFPs space, comes close enough to a fixed point of the map; which is, a point which gets mapped onto itself. Explicitly, we consider the run over and converged whenever the following logic condition is satisfied for each component of $\mathbf{v}$:

$$\forall v_j \quad : \quad |v_j - \mathrm{A}_j(\mathbf{v})| \leq \Delta v_j$$

being $\mathrm{A}(\mathbf{v})$ the algorithmic image of the HFP vector $\mathbf{v}$, and A_j its j-th component. The specification of Δv_j is clearly arbitrary. We are dealing in general with quantities of order 10^{-1}, so convergence values of order $10^{-3} \div 10^{-4}$ will be generally sufficient.

When the acceptance condition is not satisfied, which is, when at least one component of $\mathbf{v}$ is mapped by the HF algorithm to a new image value for a total displacement larger than the respective component of $\Delta\mathbf{v}$, the step is rejected and iteration continues. This lets us define the mixing parameter $g \in [0, 1]$, which represents the deterministic fractional combination of the current step with its image value to obtain a new step,

$$\mathbf{v}_{i+1} \equiv g\mathrm{A}\left(\mathbf{v}_i\right) + (1 - g)\mathbf{v}_i \tag{2.16}$$

and becomes a simple but useful knob to tune algorithmic evolution. Its usefulness comes from the core concept that, even if physically there must exist a unique solution to the model (thus just one self-consistent HFP vector, eventually of all zeros) such a fixed point could be hard to reach following a pure iterative numeric scheme on a given map (which would be the $g = 1$ case of the above equation) for a variety of reasons. By moving "slower" (low g) with an accurate optimization of the mixing, as will be seen concretely in Sec. **??**, it is possible to reach convergence in a controlled way.

When it comes to HF iterative schemes of this kind, it can happen with some frequency that the algorithm remains stuck in an infinite loop between a set of points mathematically linked by the self-consistent map but not representing a physical solution (because of their reciprocal absolute distance in HFP space). In some other situations, evolutions could turn out to be rather slow. Because of this it is reasonable to define a parameter p, such that the number of iterations is kept $i \leq p$. If the algorithm failed to converge in p steps, the point is saved as "`NaN`" (Not a number) and the algorithm moves to another simulation. Introducing this simple rule, runtime is kept under control and we gain an easy method to identify the regions of parameter space requiring more care – just read from the raw data where `NaN`s have been saved.

Chapter 3

Mean-field theory discussion of the EHM

[Introduction: this is a technical chapter about Hartree Fock applied to EHM.]

3.1 Extended Hubbard model symmetries and Wick's theorem

The general aim of this project is to study the phase diagram of the EHM by comparing ground-state energies of different phases. The phases we consider in next chapters for the EHM are the normal state (N), the antiferromagnetic ordering (AF), given by a non-uniform distribution of charge in each spin sector, and the anisotropic superconducting (SC) phase, described by a uniformly distributed charge allowing for Cooper pairing instabilities.

At the end of this section, in Tab. 3.2 we summarize all symmetries with the relative operations. Throughout the entire thesis we will refer to specific SSB for given groups to establish a specific phase.

3.1.1 Spatial and topological symmetries

Let us start from the trivial topology of the lattice we are working on. The EHM of Eq. (1.2) is evidently translational invariant on both the x and y directions. We refer to these groups as $\mathrm{T}_1^{(x)}$, $\mathrm{T}_1^{(x)}$. The "1" subscript indicates the one-site translations under which the model is symmetric.

Evidently, the following statement holds:

$$\mathrm{T}_n^{(\ell)} \supset \mathrm{T}_{kn}^{(\ell)} \qquad k \in \mathbb{N}$$

A model which is invariant for n-sites translations in the ℓ direction, also is for kn-sites translations. Then we can break translation symmetry in a given direction reducing it to a smaller group. For example, the commensurate AF phase we analyze halves translational invariance,

$$\bigotimes_{\ell=x,y} \mathrm{T}_1^{(\ell)} \overset{\mathrm{SSB}}{\rightarrow} \bigotimes_{\ell=x,y} \mathrm{T}_2^{(\ell)}$$

since it deals with two-site periodic spin-density waves (SDWs); instead the SC phase preserves translational invariance.

Another natural symmetry of the EHM is the four-fold point group C_4 of $\pi/2$ rotations, expressing the square lattice property being mapped onto itself under said rotations. A similar property as for the translations holds for point groups,

$$\mathrm{C}_n \supset \mathrm{C}_{n/k} \qquad k, n/k \in \mathbb{N}$$

which expresses the fact that, if a system is topologically invariant under n-fold rotations and $k \in \mathbb{N}$ exists such that $n/k \in \mathbb{R}$, then the smaller point group is automatically included. As for translations, we can break the point group reducing it to a smaller subgroup, say

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\rightarrow} \mathrm{C}_2$$

Phase	SSBs
AF	$\mathrm{T}_1^{(\ell)} \to \mathrm{T}_2^{(\ell)}$
	$\mathrm{SU}(2) \to \mathrm{U}^z(1)$
SC	$\mathrm{U}^c(1) \to 0$

Table 3.1 | SSB interpretation of the AF and SC phases. In the last line, the 0 on the SSB right-hand side indicates a completely disrupted symmetry.

This is what happens in general with the effective arising of anisotropic corrections to the kinetic hamiltonian due to non-local interactions, as will be discussed in Sec. **??**. For our purposes, we will leave C_4 unbroken both in the AF and SC phases.

Finally, the system also is inversion-symmetric: let I_2 be the inversion group on the two-dimensional square lattice, containing trivially just the inversion operator and the identity. In general, rotations and mirroring are different operations from a group-theoretical point of view. However, when it comes to the square lattice, operatively it holds

$$\mathrm{I}_2 \equiv \mathrm{C}_2$$

To rotate by π is the same as to mirror the lattice. The inversion symmetry is then already included in C_4, and we leave it unspecified.

3.1.2 Spin and charge symmetries

When it comes to Fermi algebra and operators, a large symmetry of the EHM is a global U(2) symmetry [2] given by the transformation

$$\hat{c}_{i\sigma} \to \sum_{\sigma'} \mathcal{U}_{\sigma\sigma'} \hat{c}_{i\sigma'} \quad \text{with} \quad \mathcal{U}_{\sigma\sigma'} \in \mathrm{U}(2) \tag{3.1}$$

The proof is trivial and follows from the fact that $\mathcal{U}^\dagger \mathcal{U} = \mathcal{U}\mathcal{U}^\dagger = \mathbb{I}$ [add proof?]. Note that the the following decomposition holds,

$$\mathrm{U}(2) = \mathrm{SU}(2) \otimes \mathrm{U}^c(1) \tag{3.2}$$

which expresses the separate charge conservation and spin rotational symmetries. In detail:

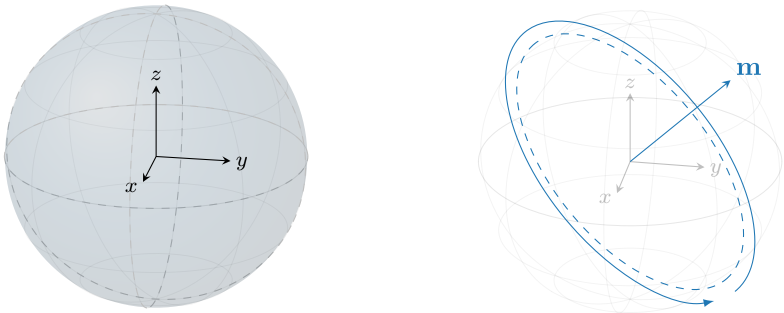
- $\mathrm{U}^c(1)$ charge conservation, coming from Eq. (3.2) decomposition. By breaking this symmetry, we allow for the total charge to fluctuate in the ground state. These fluctuations are only allowed in the SC phase in the present text;

- SU(2) spin rotation symmetry, also , coming from Eq. (3.2) decomposition. Note that:

  $$\mathrm{SU}(2) \supset \mathrm{U}^z(1)$$

  The z subscript we included to distinguish from the charge group. Hence SU(2) symmetry can be broken, selecting a particular vectorial direction for the order parameter (we call z), and reduced to a smaller $\mathrm{U}^z(1)$ symmetry which is essentially expressed by the conservation of the magnitude of the order parameter.

The last point is important: for the AF phase, for example, there is no need for the magnetization vector to be directed along the physical z-axis, which is the one perpendicular to the lattice plane. SU(2) symmetry breaks when (staggered) magnetization is established along a particular direction, but spin rotations around said direction still are ground state symmetries. This is shown schematically in Fig. 3.1.



a | SU(2) rotations domain before SSB.

b | U(1) rotations domain after SSB.

Figure 3.1 | Graphic representation of the concept of SU(2) $\to$ U^z(1) SSB. The solid color identifies, in both cases, the rotations domain for the current symmetry. When SSB takes place in a magnetic phase, the magnetization order parameter acquires a precise direction which was previously invariant over the full spherical domain. After SSB, the hamiltonian is rotationally invariant only for rotations *around* the selected magnetization vector.

3.1.3 Particle-hole symmetry at half-filling

The half-filled EHM is of particular interest due to its particle-hole symmetry (PHS), which is immediately distrupted at infinitesimal doping. Let us prove that. Consider the generic unitary operation

$$\mathrm{P} \quad : \quad \hat{c}_{i\sigma} \quad \to \quad e^{i\eta_{i\sigma}}\hat{h}^\dagger_{i\sigma}$$

being in general η_i a phase dependent on the fermionic quantum state $(i\sigma)$. Inserting the above transformation in the EHM, we get

$$\mathrm{P}\{\hat{H}\} = -t\sum_{\langle ij\rangle}\sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma})}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma} + U\sum_i \hat{h}_{i\uparrow}\hat{h}_{i\downarrow}\hat{h}^\dagger_{i\downarrow}\hat{h}^\dagger_{i\uparrow} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'}\hat{h}^\dagger_{i\sigma}$$

For the two interactions $\hat{H}_U$, $\hat{H}_V$ the phase factors cancelled out. We aim to find the conditions under which the above hamiltonian is mapped onto the EHM of (1.2),

$$\mathrm{P}\{\hat{H}\} \stackrel{?}{=} -t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma} + U\sum_i \hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma}$$

Before starting the derivation, which is an extension of the argument of Arovas et al. [2] about the conventional HM, let us discuss some properties of the hole operators. Due to the presence of the phase factors in the PHS operative definition, the Fermi anticommutation rules hold regardlessly of the phase factor:

$$\begin{aligned}\{\hat{h}_{i\sigma},\hat{h}^\dagger_{j\sigma'}\} &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})}\{\hat{c}^\dagger_{i\sigma},\hat{c}_{j\sigma'}\}\\ &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})}\delta_{ij}\delta_{\sigma\sigma'}\\ &= \delta_{ij}\delta_{\sigma\sigma'}\end{aligned}$$

and evidently:

$$\hat{h}_{i\sigma}\hat{h}^\dagger_{i\sigma} = \hat{n}_{i\sigma}$$

being $\hat{n}_{i\sigma}$ the fermionic number operator on site i with spin σ. It follows, for quartic interactions

$$\begin{aligned}
\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'}\hat{h}^\dagger_{i\sigma} &= -\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'} \\
&= \hat{h}_{i\sigma}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= -\hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= -\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \left(\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'} \\
&= \hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} + (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) - 1 - \left(\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'}
\end{aligned}$$

The first term of the last passage is mirrored with respect to the starting term. Now, let us deal with the kinetic, local and non-local interactions separately.

Kinetic term. Consider the kinetic term,

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma})}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma}$$

It suffices to anticommute the two fermionic operators, and exchange site labels $i \leftrightarrow j$, to get

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma}+\pi)}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma}$$

A necessary condition on the unitary operation P is the following,

$$\eta_{i\sigma} + \eta_{j\sigma} = (2k-1)\pi \quad \text{with} \quad k \in \mathbb{Z}$$

valid for NN sites.

Local repulsion. Substitute in $\hat{H}_U$ the above derivation,

$$\begin{aligned}
\mathrm{P}\{\hat{H}_U\} &= U\sum_i \hat{h}_{i\uparrow}\hat{h}_{i\downarrow}\hat{h}^\dagger_{i\downarrow}\hat{h}^\dagger_{i\uparrow} \\
&= U\sum_i \hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U\sum_i(\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}) - U\sum_i(1) \\
&= U\sum_i \hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U(\hat{N} - L_xL_y)
\end{aligned}$$

Thus, being for the half-filled state

$$\langle\hat{N}\rangle = L_xL_y$$

preserves PHS in the local sector.

Non-local attraction. Proceed identically for $\hat{H}_V$,

$$\begin{aligned}
\mathrm{P}\{\hat{H}_V\} &= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'} \hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'}\hat{h}^\dagger_{i\sigma} \\
&= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'} \hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}(\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) + V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}(1) \\
&= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'} \hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} - 2zV(\hat{N} - L_xL_y)
\end{aligned}$$

with $z = 4$ the lattice coordination number. In the last passage, we used that on the square lattice

$$\begin{aligned}
\sum_{\langle ij\rangle}(1) &= \sum_{i\in\mathcal{S}/2}\sum_{j\in\mathrm{NN}(i)}(1) \\
&= \frac{L_xL_y}{2} \times z
\end{aligned} \tag{3.3}$$

Symmetry	Operations
$\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$	One-site discrete translations separately in both directions
C_4	Four-fold point group (real $\pi/2$ angle rotations)
$\mathrm{U}^c(1)$	Charge-conserving global phase shifts
SU(2)	Three-dimensional rotations in spin space
PHS	Particle-hole exchange (only at half filling)
$\mathrm{U}^z(1)$	Reduced two-dimensional rotations in spin space

Table 3.2 Model symmetries and relative group operations. Note that $\mathrm{U}^c(1)$ represents the charge conserving group, while $\mathrm{U}^z(1)$ represents the subgroup obtained by breaking the SU(2) symmetry with a vector order parameter and preserving its amplitude. Since the latter is "derived" from SU(2), we separate it from the four main symmetries of the model.

which will turn out quite often. As before, an half-filled phase preserves PHS in the non-local sector. In last passage we used the fact that summing over NNs means overcounting each site $z/2$ times, and spin degeneracy.

No other condition on phases is derived, thus we conclude the half-filled EHM preserves PHS if the two sublattices $\mathcal{S}_a$, $\mathcal{S}_b$ the square lattice can be divided into, being a (trivial) bipartite lattice, differ by a π phase difference under PHS transform. In other words, if P is defined by the unitary operation

$$\mathrm{P} \quad : \quad \begin{cases} \hat{c}_{i\sigma} \to \hat{h}^\dagger_{i\sigma} & \text{for} \quad i \in \mathcal{S}_a \\ \hat{c}_{i\sigma} \to -\hat{h}^\dagger_{i\sigma} & \text{for} \quad i \in \mathcal{S}_b \end{cases}$$

and the system is half-filled, PHS is respected. Any other phase choice, as long as it alternates by π on the two sublattices, is good enough. To summarize, we report in Tab. 3.2 all identified symmetries.

3.2 MFT on Hubbard lattices

This section serves as groundwork for everything that follows. We here define the MFT scheme as well as the iterative HF computational scheme we use to estimate parametrically the relevant physical quantities of the various phases. We start by recalling the general features of MFT and then move to the specificity of the EHM, apply Wick's theorem and finally move to algorithic aspects.

3.2.1 MFT general procedure, self-consistency and Wick's theorem

The Mean Field Theory approach to any fermionic quartic operator is given by the following procedure: given the hamiltonian made of the quadratic operators $\hat{A}$ and $\hat{B}$

$$\hat{H} \equiv \hat{A}\hat{B}$$

we define the fluctuations

$$\delta\hat{A} \equiv \hat{A} - \langle\hat{A}\rangle \qquad \delta\hat{B} \equiv \hat{B} - \langle\hat{B}\rangle$$

thus the following chain holds:

$$\begin{aligned} \hat{A}\hat{B} &= \langle\hat{A}\rangle\langle\hat{B}\rangle + \delta\hat{A}\langle\hat{B}\rangle + \langle\hat{A}\rangle\delta\hat{B} + \delta\hat{A}\delta\hat{B} \\ &= -\langle\hat{A}\rangle\langle\hat{B}\rangle + \hat{A}\langle\hat{B}\rangle + \langle\hat{A}\rangle\hat{B} + \underbrace{\delta\hat{A}\delta\hat{B}}_{\mathcal{O}(\delta)^2} \end{aligned}$$

which implies

$$\langle\hat{H}\rangle = \langle\hat{A}\rangle\langle\hat{B}\rangle + \langle\delta\hat{A}\delta\hat{B}\rangle$$

Thus, if $|\langle\hat{A}\rangle\langle\hat{B}\rangle| \gg |\langle\delta\hat{A}\delta\hat{B}\rangle|$, in a statistical sense we can neglect the $\mathcal{O}(\delta)^2$ part of the hamiltonian and approximate

$$\hat{H} \simeq \hat{A}\langle\hat{B}\rangle + \langle\hat{A}\rangle\hat{B} - \langle\hat{A}\rangle\langle\hat{B}\rangle \qquad \Longrightarrow \qquad \langle\hat{H}\rangle \simeq \langle\hat{A}\rangle\langle\hat{B}\rangle \tag{3.4}$$

Term	Normal phase	Antiferromagnet	Superconductor
Hartree			
Fock	Prohibited	Prohibited	Prohibited
Cooper	Prohibited	Prohibited	

Table 3.3 Summary of Wick's terms selection rules and relevant contributions to the effective hamiltonian for each phase limitedly to $\hat{H}_U$.

Self consistency of MFT procedure is thus embodied by the condition

$$\left|\frac{\langle\delta\hat{A}\delta\hat{B}\rangle}{\langle\hat{A}\rangle\langle\hat{B}\rangle}\right| \ll 1 \qquad \Longrightarrow \qquad \left|\frac{\langle\hat{A}\hat{B}\rangle}{\langle\hat{A}\rangle\langle\hat{B}\rangle} - 1\right| \ll 1 \tag{3.5}$$

which essentially states that the least the fluctuations, the better the approximation. This rule of thumb is necessary in order to interpret the physical relevance of numerical MFT results. We will come back to this in the end, for now assuming the validity of MFT approximation and deriving the mathematical results.

3.2.2 MFT treatment of the pure Hubbard model

The pure Hubbard model, $\hat{H}_t + \hat{H}_U$, offers a simple framework for MFT analysis of the AF phase: this is described in detail in App. **??**. The key passage is there given by the MFT approximation

$$\hat{n}_{i\uparrow}\hat{n}_{i\downarrow} \simeq \hat{n}_{i\uparrow}\langle\hat{n}_{i\downarrow}\rangle + \langle\hat{n}_{i\uparrow}\rangle\hat{n}_{i\downarrow} + (\text{constants}) \tag{3.6}$$

from which the AF structure is simply recovered. In terms of the discussion of the above paragraph, we have $\hat{A} \equiv \hat{n}_{i\uparrow}$ and $\hat{B} \equiv \hat{n}_{i\downarrow}$. However the "full" operator is given by

$$\hat{n}_{i\uparrow}\hat{n}_{i\downarrow} = \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\hat{c}_{i\uparrow}$$

whose terms we can contract in three different ways following Wick's Theorem:

$$\langle\hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\hat{c}_{i\uparrow}\rangle \simeq \underbrace{\langle\hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}\rangle\langle\hat{c}_{i\downarrow}\hat{c}_{i\uparrow}\rangle}_{\text{Cooper}} - \underbrace{\langle\hat{c}^\dagger_{i\uparrow}\hat{c}_{i\downarrow}\rangle\langle\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle\hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow}\rangle\langle\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\rangle}_{\text{Hartree}} \tag{3.7}$$

where the $-$ sign in the Fock term comes from fermionic anticommutation rules.

As a first approximation, the theorem is assumed to hold (which, in a BCS-like fashion, is equivalent to assuming for the ground-state to be a coherent state). The AF ground state reduces translational invariance and rotational symmetry, as explained in Tab. 3.1. Thus, of the three terms above, when considering the pure Hubbard model in AF phase:

- Cooper fluctuations are suppressed, because they break charge conservation;
- Similarly the Fock term is null as well because if $i = j$ and $\sigma' = \overline{\sigma}$ (as is for the local interaction, which contains the operator $\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}$) the expectation values involved are describing a process breaking the survivor $\mathrm{U}^z(1)$ spin symmetry.

Thus, correctly, the Wick's decomposition of pure HM only involves Hartree-terms of Eq. (3.7), getting us back to (3.6). The distinction between the normal phase and the AF phase is given by whether the Hartree fluctuations break or not translational symmetry: AF is indeed a specific SDW ordering which weakens the latter. In general the three parts of the decomposition (Hartree, Fock, Cooper) need to be considered altogether: this is what is done in the next chapters. What said here extends naturally to the EHM: the relevant contributions for each phase given by the local repulsion $\hat{U}$ in the EHM are listed in Tab. 3.3.

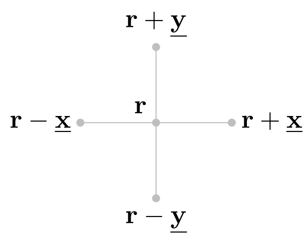


Figure 3.2 | Schematic representation of the four NNs of a given site i for a planar square lattice.

3.3 The non-local attraction $\hat{H}_V$ as a source of symmetry-breaking interactions

Let us now extend the discussion to the EHM and consider the NN non-local attraction,

$$\hat{H}_V \equiv -V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma\sigma'} \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}'\sigma'} \tag{3.8}$$

where we changed notation, and replaced the site index i with its 2D position on the lattice

$$i \quad \rightarrow \quad \mathbf{r} = (x, y)$$

Let also the lattice versors,

$$\underline{\mathbf{x}} \equiv (1, 0) \qquad \underline{\mathbf{y}} \equiv (0, 1)$$

as in Fig. 3.2.

Evidently the hamiltonian can be decomposed in various spin terms,

$$\begin{aligned}\hat{H}_V &= \sum_{\sigma\sigma'} \hat{H}_V^{\sigma\sigma'} \\ &= \underbrace{\hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow}}_{\text{Same-spin}} + \underbrace{\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow}}_{\text{Opposite-spin}}\end{aligned} \tag{3.9}$$

The role of said terms is crucial in establishing the pairing channel of the dominant physical phase. As an example, the s.s. terms can be seen as a possible source of triplet pairing superconductivity. Next section are devoted to analyze said terms and derive a simple analytical result in reciprocal space.

3.3.1 Real space description

Let us start from the real space form of Eq. (3.9). To carry out a summation over nearest neighbors $\langle \mathbf{r}\mathbf{r}' \rangle$ of a square lattice means precisely to sum over all links of the lattice. Then we can identify the generic opposite-spin (o.s.) term $\hat{H}_V^{\sigma\overline{\sigma}}$ as the one collecting the σ operators of sublattice $\mathcal{S}_a$ and $\overline{\sigma}$ operators of sublattice $\mathcal{S}_b$. The o.s. non-local interactions can be written as a sum of terms over just one of the two sublattices $\mathcal{S}_a$ and $\mathcal{S}_b$. Define the site hamiltonian

$$\hat{h}_V^{(\mathbf{r})} \equiv -V \sum_{\boldsymbol{\delta}} \left(\hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow} \right) \quad \text{with} \quad \boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\} \tag{3.10}$$

Then, it follows,

$$\begin{aligned}\hat{H}_V^{(\text{o.s.})} &= \overbrace{\sum_{\mathbf{r}\in\mathcal{S}_a} \hat{h}_V^{(\mathbf{r})}}^{\hat{H}_V^{\uparrow\downarrow}} + \overbrace{\sum_{\mathbf{r}\in\mathcal{S}_b} \hat{h}_V^{(\mathbf{r})}}^{\hat{H}_V^{\downarrow\uparrow}} \\ &= \sum_{\mathbf{r}\in\mathcal{S}} \hat{h}_V^{(\mathbf{r})}\end{aligned} \tag{3.11}$$

Sector	Term	Normal phase	Antiferromagnet	Superconductor
o.s.	Hartree			
	Fock	Prohibited	Prohibited	Prohibited
	Cooper	Prohibited	Prohibited	
s.s.	Hartree			
	Fock			
	Cooper	Prohibited	Prohibited	

Table 3.4 Summary of Wick's terms selection rules and relevant contributions to the effective hamiltonian for each phase limitedly to $\hat{H}_V$, assuming preservation of $\mathrm{U}^z(1)$ symmetry.

Here the notation of Fig. 1.1b is used. For each site $\mathbf{r}$ in a given sublattice, the nearest neighbors sites are four – all in the other sublattice. Similarly, the same-spin (s.s.) hamiltonian decomposes as

$$\hat{H}_V^{(\text{s.s.})} = -V \sum_{\mathbf{r}\in\mathcal{S}_a} \sum_{\boldsymbol{\delta},\sigma} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma}\right) \quad \text{with} \quad \boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}$$

Opposite-spin terms. Consider first the o.s. terms of Eq. (3.10): take e.g. the NNs term $\hat{n}_{i\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}$. As in Eq. (3.7), taking Wick's contractions,

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \\ &\simeq \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle}_{\text{Hartree}}\end{aligned}$$

Identical decompositions are given for all others NNs. Of the three terms above:

- The Cooper term breaks $\mathrm{U}^c(1)$ charge symmetry, allowing for supeconducting instabilities;
- The Fock term breaks $\mathrm{U}^z(1)$ symmetry, because it accounts for a site hop *plus* spin flip process;
- The Hartree term can reduce both translational invariance and spin rotational symmetry.

As a general rule of thumb, throughout this text we will discard the o.s. Fock term eveywhere, preserving a robust $\mathrm{U}^z(1)$ symmetry. For the normal state as well as for the AF instability only the Hartree term is to be accounted; for the SC phase also the Cooper term contributes. Note that for superconducting instabilities, due to superexchange mechanism (as explained in App. **??**) the o.s. term accounts for singlet pairing as well as zero-projection triplet pairing. Which channel is preferred, is a matter of thermodynamic advantage.

Same-spin terms. Consider then the same-spin terms: take e.g. the term $\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}$. As above,

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \\ &= \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle\langle \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle}_{\text{Hartree}}\end{aligned}$$

Identical consideration as in the above paragraph hold for each term. The only difference with the o.s. terms is given by the Fock term: since the spin-flip process is absent, now the Fock fluctuations actually contribute as an effective NN hopping term. In other words, this term does not break $\mathrm{U}^z(1)$ symmetry and thus is perfectly legitimate in each phase. Tab. 3.4 summarizes the selection rules hereby presented.

3.3.2 Fourier transform of the model

It is useful to derive analytically the reciprocal-space form of the three hamiltonian parts. By the sign $\overset{\mathcal{F}}{=}$, we indicate a passage where a Fourier-transform is performed. $\mathcal{F}$-transform is obtained according to the convention Fourier transform it according to the convention

$$\hat{c}_{\mathbf{r}\sigma} \overset{\mathcal{F}}{=} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in\mathrm{BZ}} e^{-i\mathbf{k}\cdot\mathbf{r}} \hat{c}_{\mathbf{k}\sigma} \tag{3.12}$$

Kinetic part. The kinetic part is the easiest to transform. Apply the definition,

$$\begin{aligned}
-t\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}'\sigma}+\text{h.c.}\right] &\overset{\mathcal{F}}{=} -t\sum_{\langle\mathbf{r}\mathbf{r}'\rangle}\sum_{\sigma}\frac{1}{\sqrt{L_xL_y}}\sum_{\mathbf{k}\in\mathrm{BZ}}e^{i\mathbf{k}\cdot\mathbf{r}}\hat{c}^\dagger_{\mathbf{k}\sigma}\frac{1}{\sqrt{L_xL_y}}\sum_{\mathbf{k}'\in\mathrm{BZ}}e^{-i\mathbf{k}'\cdot\mathbf{r}'}\hat{c}_{\mathbf{k}'\sigma}+\text{h.c.}\\
&= -2t\sum_{\mathbf{k},\mathbf{k}'}\sum_{\sigma}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma}\frac{1}{L_xL_y}\sum_{\mathbf{r}\in\mathcal{S}/2}e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}}\left[e^{ik'_x}+e^{-ik'_x}+e^{ik'_y}+e^{-ik'_y}\right]\\
&= \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\sigma}\epsilon_{\mathbf{k}}\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}
\end{aligned} \tag{3.13}$$

where the bare bands were defined,

$$\epsilon_{\mathbf{k}} \equiv -2t\left(\cos k_x + \cos k_y\right) \tag{3.14}$$

Non-local attraction. Consider a generic bond, say, the one connecting NNs $\mathbf{r}$ and $\mathbf{r}\pm\boldsymbol{\delta}$. Then:

$$\begin{aligned}
\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'} &= \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\sigma}\\
&\overset{\mathcal{F}}{=} \frac{1}{(L_xL_y)^2}\sum_{\mathbf{k}_1,\cdots,\mathbf{k}_4}e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}}e^{\pm i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}\hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}
\end{aligned}$$

Then, the interaction at site $\mathbf{r}$, spin σ with its NNs at spin σ' – indicated as $(\mathbf{r}\sigma\sigma')$ – is given by

$$\begin{aligned}
(\mathbf{r}\sigma\sigma') &= -V\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}}\left[\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}+\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'}\right]\\
&\overset{\mathcal{F}}{=} -\frac{V}{(L_xL_y)^2}\sum_{\boldsymbol{\delta}}\sum_{\mathbf{k}_1,\cdots,\mathbf{k}_4}e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}}\\
&\qquad\qquad\times\left(e^{i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}+e^{-i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}\right)\hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}\\
&= -\frac{2V}{(L_xL_y)^2}\sum_{\boldsymbol{\delta}}\sum_{\mathbf{k}_1,\cdots,\mathbf{k}_4}e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}}\cos\left[(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}\right]\hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}
\end{aligned}$$

The full non-local interaction is given by summing over all sites of $\mathcal{S}/2$ (which is, one sublattice of $\mathcal{S}$). This gives back momentum conservation,

$$\frac{1}{L_xL_y}\sum_{\mathbf{r}\in\mathcal{S}/2}e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}$$

Let $\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4=\mathbf{K}$, and define $\mathbf{k}$, $\mathbf{k}'$ such that

$$\mathbf{k}_1\equiv\mathbf{K}+\mathbf{k}\qquad \mathbf{k}_2\equiv\mathbf{K}-\mathbf{k}\qquad \mathbf{k}_3\equiv\mathbf{K}-\mathbf{k}'\qquad \mathbf{k}_4\equiv\mathbf{K}+\mathbf{k}'\qquad \delta\mathbf{k}\equiv\mathbf{k}-\mathbf{k}'$$

Sums over these variables must be intended as over the Brillouin Zone (BZ). Hence:

$$\begin{aligned}
\hat{H}_V &= \sum_{\mathbf{r}\in\mathcal{S}_a}\sum_{\sigma\sigma'}(\mathbf{r}\sigma\sigma')\\
&\overset{\mathcal{F}}{=} -\frac{V}{L_xL_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\cos\left(\delta\mathbf{k}\cdot\boldsymbol{\delta}\right)\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\\
&= -\frac{V}{L_xL_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}
\end{aligned} \tag{3.15}$$

which decomposes in s.s. and o.s. channels as:

$$\hat{H}_V \overset{\mathcal{F}}{=} \overbrace{-\frac{V}{L_x L_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}^{\text{s.s.}}$$
$$\underbrace{-\frac{V}{L_x L_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}_{\text{o.s.}}$$

The o.s. part can be simplified: since

$$\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} = \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}$$

due to fermionic anti-commutation rules, and since the composite transform

$$\mathbf{k}\to-\mathbf{k} \qquad \mathbf{k}'\to-\mathbf{k}'$$

leaves the form factor $\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)$ invariant, we reduce the spin sum to two times the $\sigma=\uparrow$ part,

$$\hat{H}_V^{(\text{o.s.})} = -\frac{2V}{L_x L_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{3.16}$$

whiel the explicit s.s. sector is given immediately by:

$$\hat{H}_V^{(\text{s.s.})} = -\frac{V}{L_x L_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]$$
$$\times\left[\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\uparrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\uparrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow}+\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\downarrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\downarrow}\right] \tag{3.17}$$

The result here obtained will be used various times in next chapters. Note that different Wick contraction schemes will be referred to as:

$$c^\dagger_{\mathbf{K}+\mathbf{k}\sigma}c^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}c_{\mathbf{K}-\mathbf{k}'\sigma'}c_{\mathbf{K}+\mathbf{k}'\sigma} \qquad \text{Cooper contraction} \tag{3.18}$$
$$c^\dagger_{\mathbf{K}+\mathbf{k}\sigma}c^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}c_{\mathbf{K}-\mathbf{k}'\sigma'}c_{\mathbf{K}+\mathbf{k}'\sigma} \qquad \text{Fock contraction} \tag{3.19}$$
$$c^\dagger_{\mathbf{K}+\mathbf{k}\sigma}c^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}c_{\mathbf{K}-\mathbf{k}'\sigma'}c_{\mathbf{K}+\mathbf{k}'\sigma} \qquad \text{Hartree contraction} \tag{3.20}$$

When contracting in the Fock channel, a minus sign must be included.

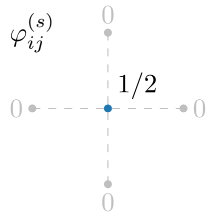
Local repulsion. A somewhat identical argument can be carried out for the local interaction,

$$\hat{H}_U = U\sum_{\mathbf{r}\in\mathcal{S}}\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\downarrow}$$

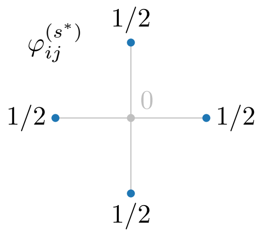
The only difference here is made by the fact that the interaction is repulsive on-site, thus the final structure factor in the Fourier transform disappears as well as the minus sign,

$$\hat{H}_U \overset{\mathcal{F}}{=} \frac{U}{L_x L_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{3.21}$$

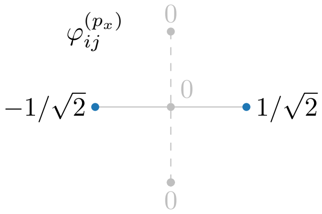
When contracting these fermionic operators, identical considerations as for the non-local counterpart (3.15) hold. Note the similarity of this Fourier transform with the o.s. sector of the non-local attraction, given in Eq. (3.15). With this, we conclude the general introduction to the model and pass to the setup of the Hartree-Fock (HF) algorithm.



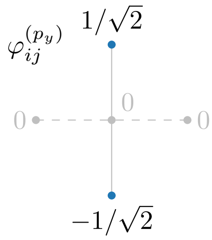
a | Local s-wave



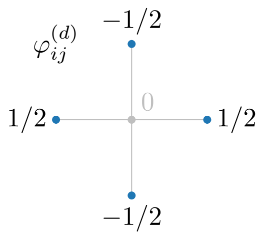
b | Extended s-wave.



c | p_x-wave.



d | p_y-wave.



e | $d_{x^2-y^2}$-wave.

Figure 3.3 | Form factors at different topologies, as listed in Tab. 3.5. In figures five sites are represented: the hub site i and its four NN. Solid lines represent non-zero values for $\varphi_{\boldsymbol{\delta}}$, while dashed lines represent vanishing factors.

3.4 Square lattice spatial symmetry channels

In next sections we introduce spatial symmetry structures and discuss their relation with point groups. We will aways refer to various spatial structures as "symmetry channels", throughout the system is able to establish a given order parameter defined in space.

Looking at $\hat{H}$, the main difference between its two-body interactions $\hat{H}_U$ and $\hat{H}_V$ of course relies in locality (first) and non-locality (second). Both operators in principle are able to break C_4 symmetry, as will later be discussed extensively.

For the 2D square lattice, various spatial symmetries are possible. Let i be a specific site, and consider a generic complex function g_{ij} whose domain is given by the site itself and its four NNs (the domain concides with the points of Fig. 3.2). The function can be written in the C_4 basis

$$g_{ij} = \sum_{\gamma} g^{(\gamma)} \varphi_{ij}^{(\gamma)}$$

where $g^{(\gamma)} \in \mathbb{C}$ are the g_{ij} symmetry channels coefficients, while $\varphi_{ij}^{(\gamma)}$ are the form factors listed in Tab. 3.5 and plotted in Fig. 3.4, a simple orthonormal rearrangement of the harmonics basis.

When dealing with phases which respect translational symmetry, or at most reduce it to a weaker symmetry by shrinking the Brillouin Zone, it's useful to work in reciprocal space, as discussed in Sec. 3.3. As said, a particular phase can show specific spatial symmetries, and so will the Hartree-Fock parameters. The key point to notice is that self-consistency equations involve sums (or integrals) to be computed over fermionic states, which can be heavily simplified by employing said symmetries. Consider the form factors of Tab. 3.5, and their Fourier transforms of Tab. 3.6. They satisfy the relations:

$$\frac{1}{L_x L_y} \sum_{ij} \varphi_{ij}^{(\gamma)} \varphi_{ij}^{(\gamma)*} = \delta_{\gamma\gamma'} \qquad \frac{1}{L_x L_y} \sum_{\mathbf{k}} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} = \delta_{\gamma\gamma'}$$

If these symmetries are developed by the HFPs, it's of no use to integrate over the entire BZ, since different regions linked by periodicity of the form factors will lead to redundant calculations.

Structure	Form factor	Graph
s-wave	$\varphi^{(s)}_{\mathbf{rr'}} = \delta_{\mathbf{rr'}}$	Fig. 3.3a
Extended s-wave	$\varphi^{(s^*)}_{\mathbf{rr'}} = \frac{1}{2}\left[\delta_{\mathbf{r'}=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r'}=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r'}=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r'}=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 3.3b
p_x-wave	$\varphi^{(p_x)}_{\mathbf{rr'}} = \frac{1}{\sqrt{2}}\left[\delta_{\mathbf{r'}=\mathbf{r}+\underline{\mathbf{x}}} - \delta_{\mathbf{r'}=\mathbf{r}-\underline{\mathbf{x}}}\right]$	Fig. 3.3c
p_y-wave	$\varphi^{(p_y)}_{\mathbf{rr'}} = \frac{1}{\sqrt{2}}\left[\delta_{\mathbf{r'}=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r'}=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 3.3d
$d_{x^2-y^2}$-wave	$\varphi^{(d)}_{\mathbf{rr'}} = \frac{1}{2}\left[\delta_{\mathbf{r'}=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r'}=\mathbf{r}-\underline{\mathbf{x}}} - \delta_{\mathbf{r'}=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r'}=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 3.3e

Table 3.5 Form factors for a function defined on a square lattice, with a prefactor included for normalization. We use lattice vector notation here. The last column indicates the graph representation of each contribution given in Fig. 3.3. Subscript $x^2 - y^2$ is omitted for notational clarity.

Structure	Structure factor	Graph
s-wave	$\varphi^{(s)}_{\mathbf{k}} = 1$	Fig. 3.3a
Extended s-wave	$\varphi^{(s^*)}_{\mathbf{k}} = \cos k_x + \cos k_y$	Fig. 3.3b
p_x-wave	$\varphi^{(p_x)}_{\mathbf{k}} = i\sqrt{2}\sin k_x$	Fig. 3.3c
p_y-wave	$\varphi^{(p_y)}_{\mathbf{k}} = i\sqrt{2}\sin k_y$	Fig. 3.3d
$d_{x^2-y^2}$-wave	$\varphi^{(d)}_{\mathbf{k}} = \cos k_x - \cos k_y$	Fig. 3.3e

Table 3.6 Structure factors derived from the correlation structures of Tab. 3.5. The functions hereby defined are orthonormal, and are plotted in Fig- 3.4.

Finally, a very useful detail is given by the following argument. Consider the equation

$$\cos(\delta k_x) + \cos(\delta k_y) = \cos k_x \cos k'_x + \sin k_x \sin k'_x + \cos k_y \cos k'_y + \sin k_y \sin k'_y$$

For the sake of readability, the notations

$$c_\ell \equiv \cos k_\ell \qquad s_\ell \equiv \sin k_\ell \qquad c'_\ell \equiv \cos k'_\ell \qquad s'_\ell \equiv \sin k'_\ell$$

are used. Group the four terms above,

$$\underbrace{\left(c_x c'_x + c_y c'_y\right)}_{\text{Symmetric}} + \underbrace{\left(s_x s'_x + s_y s'_y\right)}_{\text{Anti-symmetric}} \tag{3.22}$$

The first two exhibit inversion symmetry for both arguments $\mathbf{k}$, $\mathbf{k}'$; the second two exhibit anti-symmetry. Decoupling the symmetric part, we obtain

$$c_x c'_x + c_y c'_y = \frac{1}{2}(c_x + c_y)(c'_x + c'_y) + \frac{1}{2}(c_x - c_y)(c'_x - c'_y)$$

which finally gives:

$$\begin{aligned}
\cos(\delta k_x) + \cos(\delta k_y) = {} & \frac{1}{2}(c_x + c_y)(c'_x + c'_y) && (s^*\text{-wave}) \\
& + s_x s'_x && (p_x\text{-wave}) \\
& + s_y s'_y && (p_y\text{-wave}) \\
& + \frac{1}{2}(c_x - c_y)(c'_x - c'_y) && (d_{x^2-y^2}\text{-wave})
\end{aligned}$$

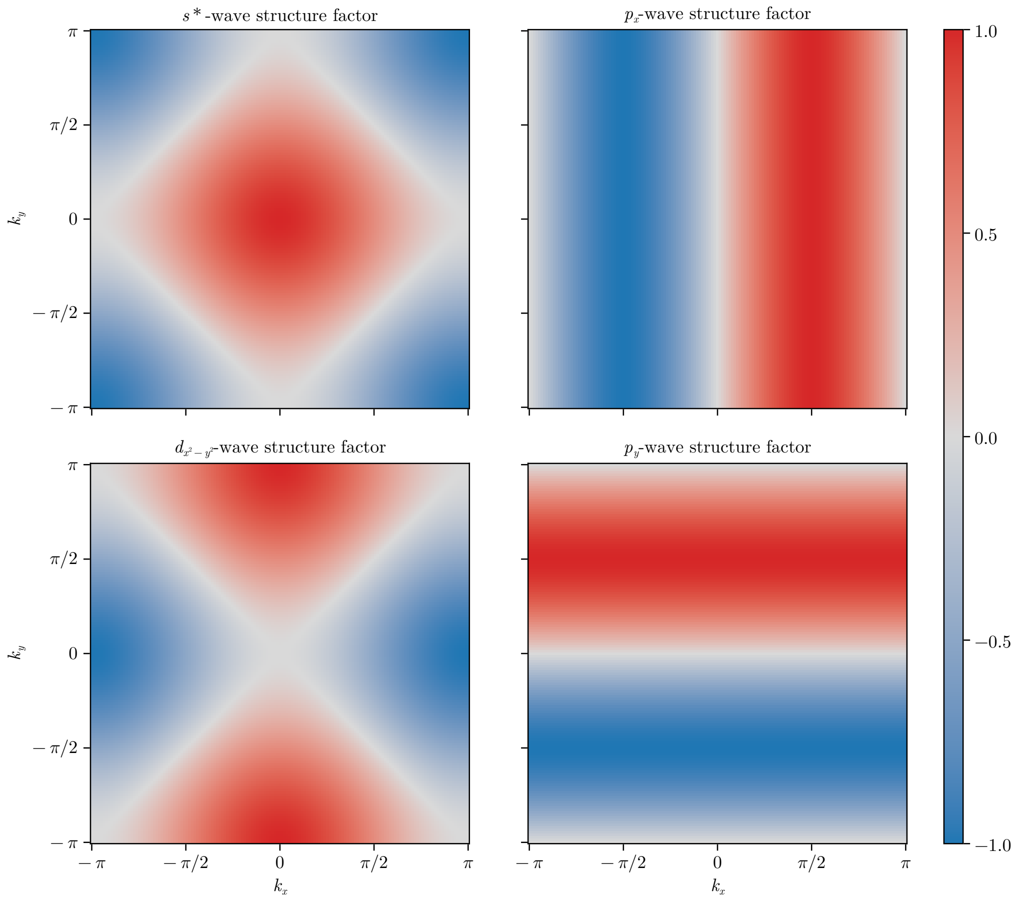


Figure 3.4 Representation of the various structure factors listed in Tab. 3.6 on the BZ. For the s^* and $d_{x^2-y^2}$-wave factors, the real part is plotted; for the p_ℓ factors, the imaginary part is plotted.

or, in a more compact form

$$\cos(\delta k_x) + \cos(\delta k_y) = \frac{1}{2}\sum_\gamma \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}'}^{(\gamma)*} \quad \text{for } \gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\} \tag{3.23}$$

$$= \frac{1}{2}\sum_\gamma \varphi_{\mathbf{k}}^{(\gamma)*} \varphi_{\mathbf{k}'}^{(\gamma)} \tag{3.24}$$

This decomposition will become particularly handy in later calculations.

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